VISHWAKARMA INSTITUTE OF TECHNOLOGY, PUNE

Department of Artificial Intelligence and Machine Learning

FEW-SHOT TEMPORAL ANOMALY GENERATION AND DETECTION IN BRAIN CONNECTIVITY SUBGRAPHS FOR AUTISM DISORDER

Branch: AIML-B

GRN: 4

Day: Tuesday

Date: 13/10/2026

Project Guide

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INTRODUCTION

The Problem

- Autism Spectrum Disorder (ASD) is associated with differences in functional brain connectivity, measurable through resting-state fMRI (rs-fMRI).
- Representing the brain as a graph - regions of interest (ROIs) as nodes, functional connectivity as weighted edges - captures relational structure a single time series cannot.
- Connectivity is not static: dividing a scan into sliding time windows yields a temporal sequence of graphs, so a purely static representation can miss dynamics that unfold over time.
- Rather than a direct disease classifier, this project asks whether unusual spatio-temporal subgraph behavior can be identified relative to dynamics learned only from healthy (control) subjects.
- Real abnormal (ASD) examples are limited relative to what a data-hungry detector needs - motivating a second question: can structure-aware synthetic anomalies support detection in this few-shot setting?

Project Framing

The project investigates whether meaningful structure-aware anomalies can be generated from learned normal brain-network structure, and whether they can subsequently support few-shot anomaly detection.

Scope note: this is a research prototype for anomaly generation and detection, not a clinical diagnostic tool. No claim of ASD diagnosis is made.

OBJECTIVES

1 **Build time-resolved brain graphs**

Convert 871 ABIDE I scans (360 regions) into sliding-window brain graphs.

✓ COMPLETED

2 **Track brain-region subgraphs**

Follow 20 selected regions and their neighbours across time windows.

✓ COMPLETED

3 **Learn normal brain dynamics**

Train a GIN + GRU on control scans only to predict the next window.

✓ COMPLETED

4 **Detect anomalies**

Score prediction errors and set thresholds from validation controls.

✓ COMPLETED

5 **Generate synthetic anomalies**

Gradient-guided edge edits on control subgraphs (7,200 records, 0 rule violations).

✓ COMPLETED

6 **Evaluate few-shot and severity**

Test 1, 5, 10 ASD scans (accuracy, F1, AUROC); severity and demo app.

✓ COMPLETED

LITERATURE SURVEY

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Theme	Key Paper	Key Insight	Relevance to Project
ABIDE dataset	A. Di Martino et al. (2014)	Openly shares rs-fMRI + phenotypic data for 539 ASD / 573 control subjects across 17 sites; reports baseline hypoconnectivity in ASD.	Grounds the ABIDE I dataset (871 subjects, 20 sites) used throughout.
GNN expressive power	Keyulu Xu et al. (2018)	Proves common GNN variants are weaker than the Weisfeiler-Lehman test; introduces GIN, which matches it.	Justifies the weighted GIN encoder used to embed each subgraph.
Dynamic functional connectivity	R. Matthew Hutchison et al. (2013)	Defines and surveys sliding-window estimation of time-varying connectivity and its methodological pitfalls.	Grounds the 30-TR / stride-10 sliding-window graph construction.
Graph anomaly detection	Xiaoxiao Ma et al. (2021)	Taxonomizes deep graph-anomaly-detection methods by anomaly type and technique; flags scarce labels as an open challenge.	Frames the prediction-error, threshold-calibrated anomaly score.
Gradient-based graph perturbation	Kaidi Xu et al. (2019)	Formulates graph structure attacks as constrained optimization solved with projected gradient descent (PGD).	Grounds the PGD-style iterative edge-weight perturbation.

Existing Work → Our Focus



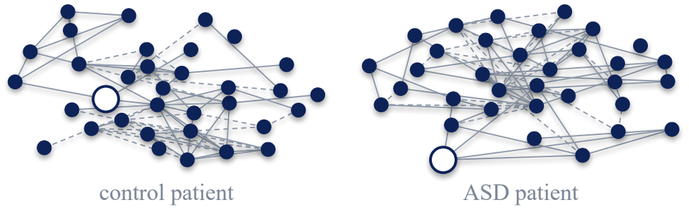
Our focus: existing work provides graph representations, temporal modeling, classification, and anomaly detection; this project additionally investigates structure-aware, gradient-guided anomaly generation for a few-shot brain-subgraph setting. Full 20-paper analysis: *GRAPH_ANOMALY_Literature_Review.xlsx*

PROPOSED METHODOLOGY

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DATA PREPARATION AND PATIENT SPLIT

ABIDE I, 871 patient scans



control patient ASD patient

SLIDING-WINDOW BRAIN GRAPH BUILDING

30-TR windows, stride 10; Pearson correlation; signed graph, top 20% edges

BRAIN-REGION SUBGRAPH TRACKING

20 selected brain regions (ROIs), 1-hop neighbourhood; node set fixed at the first window

GIN ENCODING AND GRU NEXT-WINDOW PREDICTION

2-layer weighted GIN, mean pool, 128-D embedding; GRU (hidden 128), MSE + variance loss, controls only

TEMPORAL SUBGRAPH ANOMALY SCORING FOR ASD DETECTION

patient score = max over regions and transitions

CONTROL-BASED DETECTION THRESHOLD CALIBRATION

95th percentile of validation-control scores (transition and patient level)

SYNTHETIC SUBGRAPH ANOMALY GENERATION

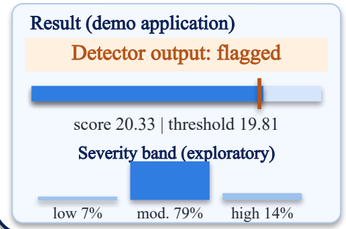
Gradient-guided PGD edge-weight edits, 5 steps, up to 15% of edge pairs, 3 severities

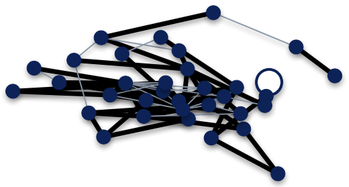
FEW-SHOT ASD ADAPTATION

1 / 5 / 10 ASD patient scans + 30 generated anomalies; logistic regression on 20-D region scores

ANOMALY DETECTION AND SEVERITY OUTPUT

Flag if score exceeds threshold; ADOS-2 severity band





synthetic anomaly (bold = edited)

MAJOR CONTRIBUTIONS

Same pipeline order as the methodology; each box states what is new at that step.

TIME-RESOLVED BRAIN GRAPHS
One signed brain network per 30-TR window instead of a single static network

→

BRAIN-REGION SUBGRAPH TRACKING
20 regions with fixed 1-hop node sets followed over time, so an anomaly is localised in region and time

→

CONTROL-ONLY TEMPORAL MODEL
GIN + GRU predict the next window from control scans only; no labelled anomalies needed for training

↓

TEMPORAL SUBGRAPH ANOMALY SCORING
Prediction error per region and step; patient score = maximum over regions and steps

←

TWO-LEVEL THRESHOLD CALIBRATION
Step and patient thresholds from validation controls; fixed a failure that flagged all 131 test patients

←

GRADIENT-GUIDED ANOMALY GENERATION
Weight-only edits of real control subgraphs; 45.8% detected against 4.8% for random edits

↓

GENERATION VALIDITY CHECKS
7,200 records, 0 rule violations, severity-monotonic, edit diversity close to 1.0

→

HONEST FEW-SHOT EVALUATION
Chance-matched F1 shows the gain comes from the flag rate, not from ranking (AUROC about 0.5)

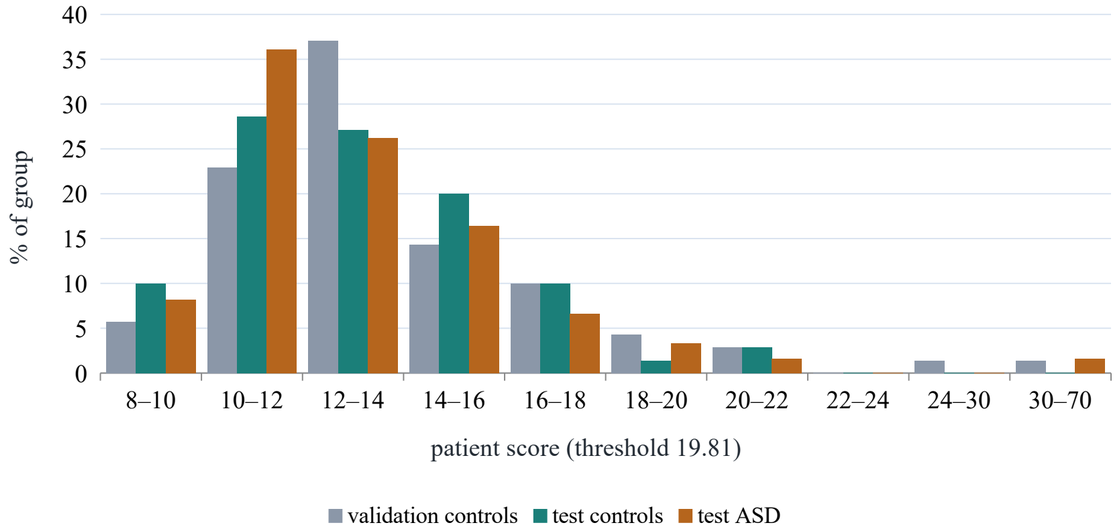
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SEVERITY ESTIMATE AND DEMO APPLICATION
ADOS-based severity bands (no better than a majority guess) and a web app for raw time-series upload

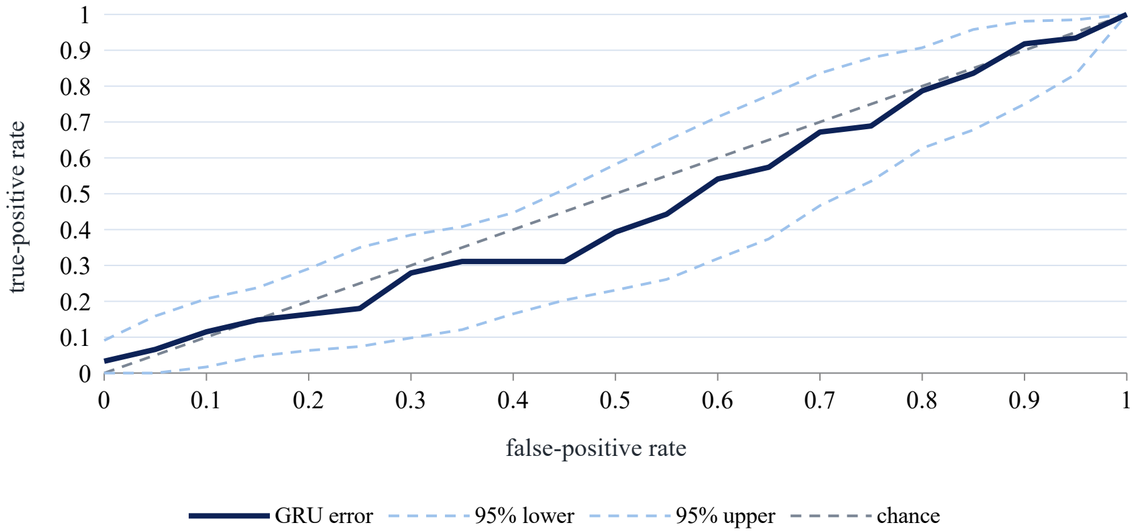
RESULTS: DETECTOR

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Patient score distribution (% of group)



ROC curve (AUROC 0.453)



Does modelling time help? Three scoring methods on the same frozen encoder (131 test patients)

Scoring method	Accuracy	Precision	Recall	F1	AUROC [95% CI]
Static (distance to control mean)	54.2%	0.556	0.082	0.143	0.452 [0.356, 0.553]
Persistence (change from previous window)	53.4%	0.500	0.033	0.062	0.485 [0.380, 0.582]
GRU prediction error (ours, the detector)	53.4%	0.500	0.033	0.062	0.453 [0.353, 0.552]

The detector flagged 4 of 131 patients; all three scores are at chance (AUROC 0.45–0.49, overlapping intervals).

RESULTS: FEW-SHOT COMPARISON

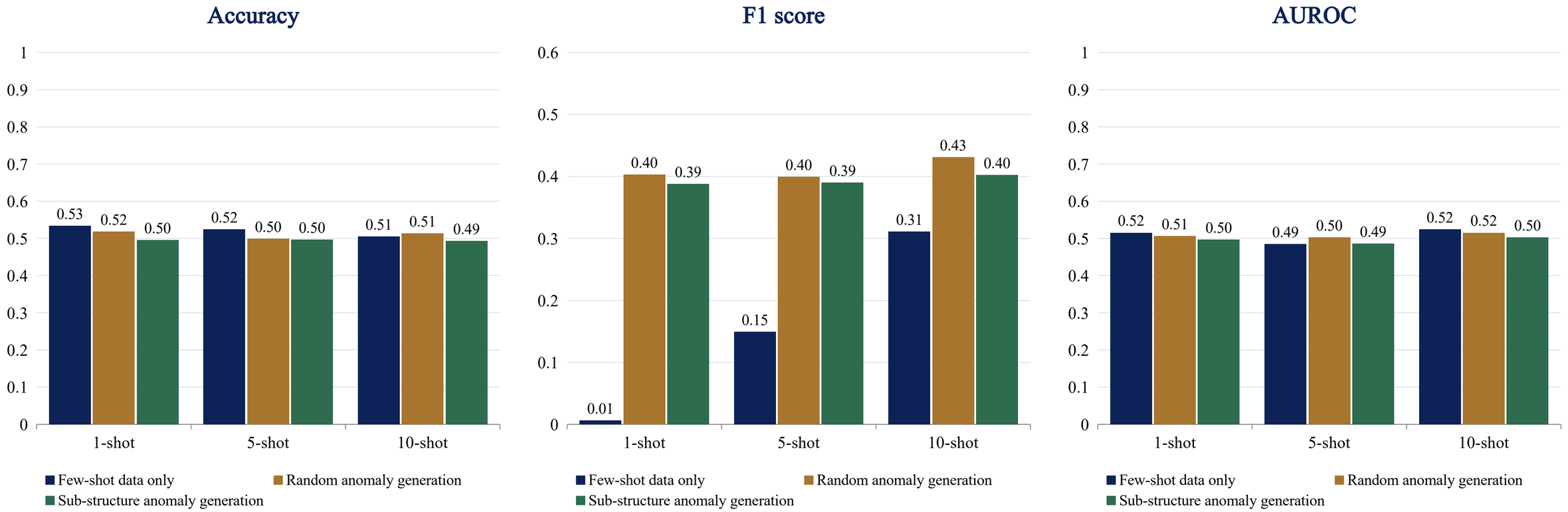
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Shots	Approach	Accuracy	Precision	Recall	F1 score	AUROC
–	Baseline detector (no few-shot data)	53.4%	0.500	0.033	0.062	0.453
1	Few-shot data only (normal detection)	**53.4% ± 0.5**	0.200 ± 0.40	0.003 ± 0.01	0.006 ± 0.01	**0.515 ± 0.06**
	Random anomaly generation + detection	51.8% ± 2.6	**0.473 ± 0.04**	**0.357 ± 0.09**	**0.403 ± 0.07**	0.506 ± 0.04
	Sub-structure anomaly generation + detection	49.5% ± 3.3	0.445 ± 0.05	0.344 ± 0.04	0.388 ± 0.04	0.497 ± 0.03
5	Few-shot data only (normal detection)	**52.4% ± 2.1**	0.430 ± 0.13	0.092 ± 0.04	0.149 ± 0.07	0.485 ± 0.02
	Random anomaly generation + detection	49.9% ± 1.7	**0.451 ± 0.03**	**0.361 ± 0.06**	**0.399 ± 0.05**	**0.502 ± 0.02**
	Sub-structure anomaly generation + detection	49.6% ± 1.4	0.447 ± 0.02	0.348 ± 0.04	0.390 ± 0.03	0.486 ± 0.02
10	Few-shot data only (normal detection)	50.5% ± 2.7	0.433 ± 0.07	0.249 ± 0.09	0.311 ± 0.09	**0.524 ± 0.05**
	Random anomaly generation + detection	**51.3% ± 2.1**	**0.474 ± 0.03**	**0.400 ± 0.06**	**0.431 ± 0.04**	0.515 ± 0.05
	Sub-structure anomaly generation + detection	49.3% ± 3.3	0.445 ± 0.05	0.367 ± 0.05	0.402 ± 0.05	0.502 ± 0.05

- **Reading the table:** mean ± SD over five support-set draws on 131 test patients (61 ASD, 70 control); green = best of the three approaches within each shot count.
- Generated anomalies raise recall and F1 (1-shot F1 0.006 → about 0.39–0.40), but AUROC stays near 0.50 and accuracy near 50%: random and sub-structure generation perform alike.
- Chance references: AUROC 0.5; accuracy 53.4% (always answer control); F1 of a random flagger at the same flag rate ≈ 0.40–0.42.

RESULTS: COMPARISON CHARTS

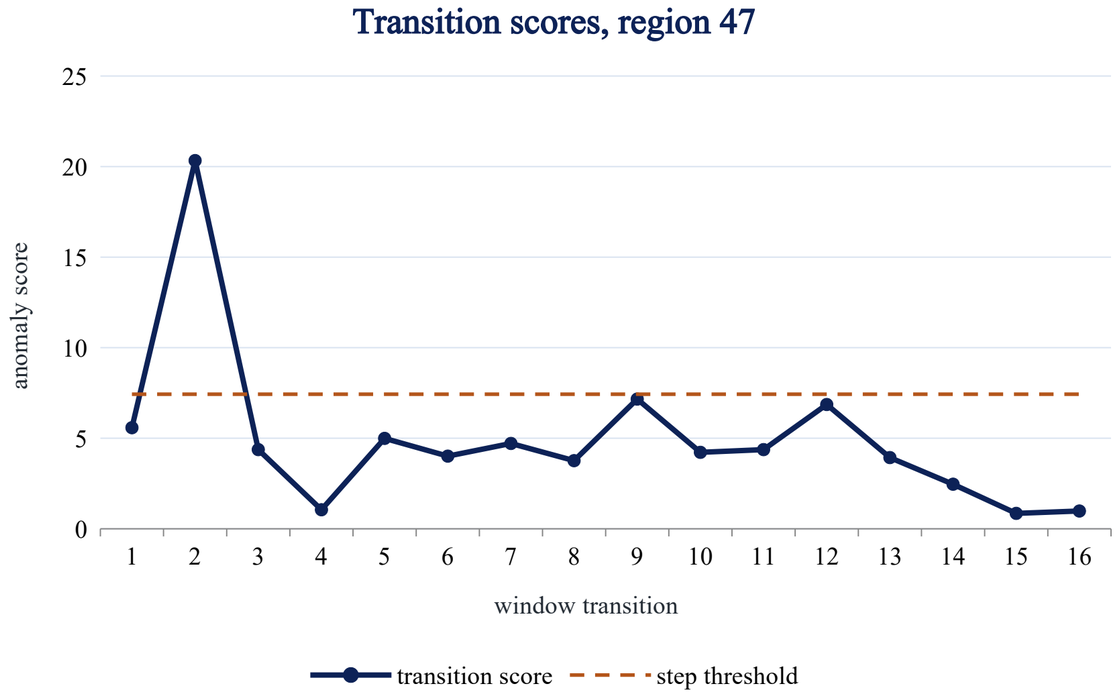
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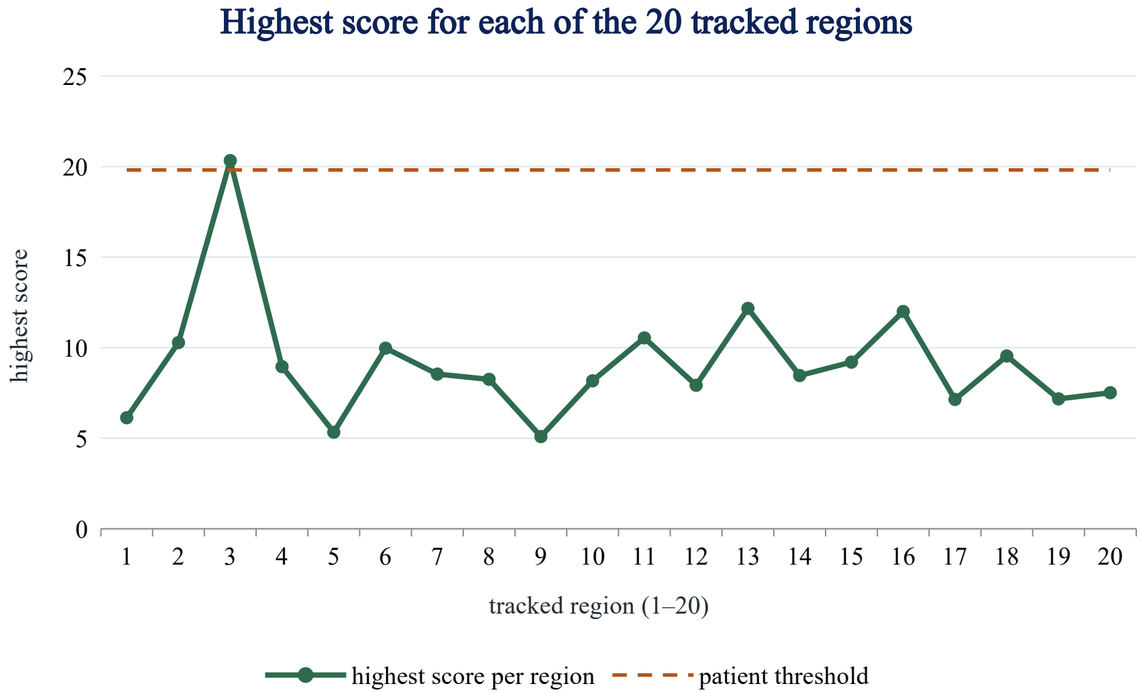


- Few-shot data only vs. random anomaly generation vs. sub-structure (gradient-guided) anomaly generation, mean of five draws.
- Anomaly generation raises F1 through a higher flag rate; AUROC and accuracy do not improve (chance: AUROC 0.5, accuracy 53.4%).
- Sub-structure generation is detected 45.8% of the time against 4.8% for random edits (generation test), yet gives no extra gain in the few-shot comparison.

RESULTS: OUTPUT FOR ONE SCAN

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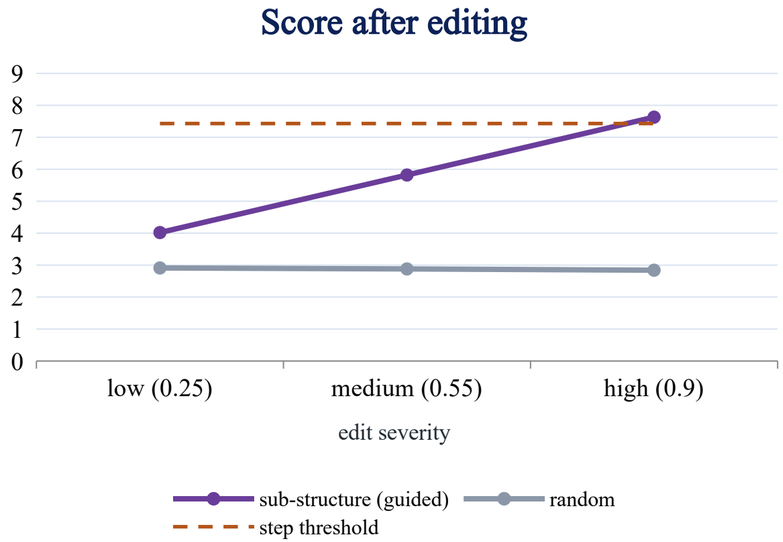


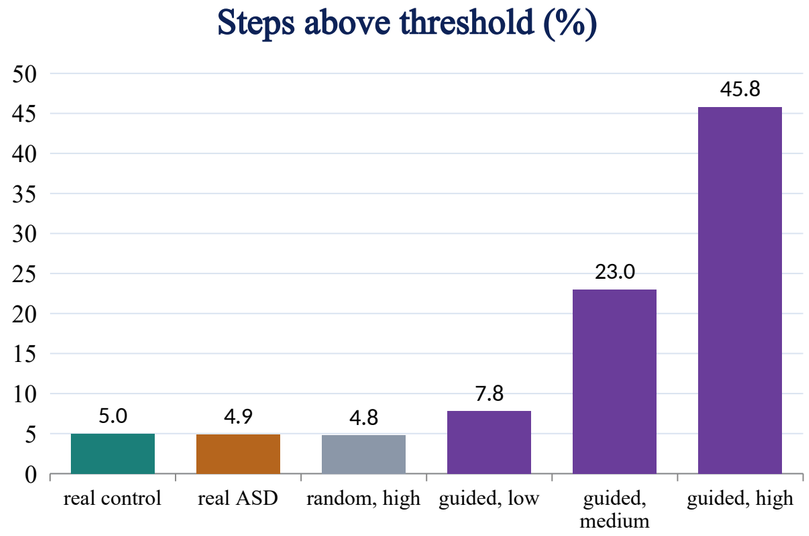


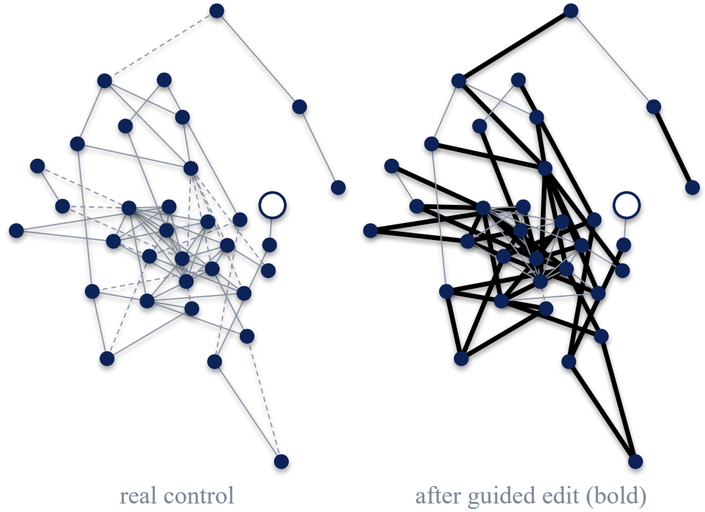
- Example: one held-out ASD test scan (196 time points, 17 windows); the scan the demo application opens.
- Patient score 20.33 against threshold 19.81: the scan is flagged; 22 of 320 transitions exceed the step threshold.
- A typical scan scores below the threshold: on the 131 test patients only 4 were flagged.

RESULTS: SYNTHETIC ANOMALIES

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Patient (control)	Method	Severity	Edge pairs edited	Score before	Score after
KKI_002	sub-structure	high	74 / 548	4.44	8.13
KKI_002	random	high	74 / 548	4.44	2.87
OHSU_001	sub-structure	medium	33 / 395	9.58	11.26
OHSU_001	random	medium	33 / 395	9.58	9.63
UCLA_1_002	sub-structure	low	20 / 526	0.56	2.64
UCLA_1_002	random	low	20 / 526	0.56	0.54

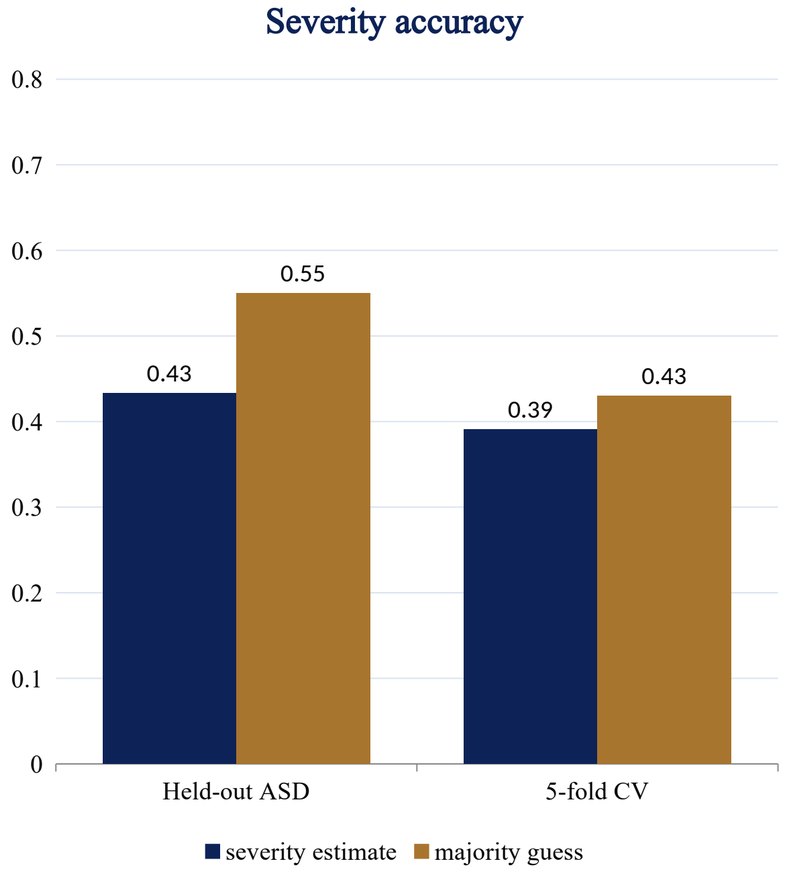
Six real generated records (each traceable to an exact subject, region and window). Drawing shows 34 nodes of one subgraph; 45 of the 60 edited edges are visible (score 0.62 → 4.83).

RESULTS: SEVERITY ESTIMATION

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Evaluation	n	Accuracy	Macro-F1	Majority guess
Held-out ASD patients	60	0.433	0.395	0.550
5-fold CV × 10	193	0.391	0.372	0.430

- Severity bands come from the ADOS-2 calibrated severity score: low (≤5), moderate (6–7), high (≥8).
- ADOS scores exist for 193 of 403 ASD patients; the estimator is trained on training patients only.
- The estimate is below a majority guess (43.3% vs 55.0%): the scores carry no usable severity signal.
- Predicted vs true severity score: Spearman 0.13 (p = 0.32).
- Severity is only estimated for scans that the detector flags.



RESULTS: DEMO APPLICATION

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File	True label	ADOS band	Score	Detector output	Detector right?
sample_01	ASD	moderate	12.583	not flagged	no
sample_02	ASD	low	11.73	not flagged	no
sample_03	Control	–	15.027	not flagged	yes
sample_04	Control	–	13.919	not flagged	yes
sample_05	Control	–	15.624	not flagged	yes
sample_06	ASD	high	14.856	not flagged	no
sample_07	ASD	moderate	15.051	not flagged	no
sample_08	Control	–	17.58	not flagged	yes
sample_09	ASD	high	8.835	not flagged	no
sample_10	Control	–	17.854	not flagged	yes
sample_11	Control	–	9.901	not flagged	yes
sample_12	Control	–	14.223	not flagged	yes
sample_13	Control	–	11.576	not flagged	yes
sample_14	ASD	high	11.016	not flagged	no
sample_15	ASD	high	8.343	not flagged	no
sample_16	ASD	high	11.868	not flagged	no
sample_17	ASD	high	15.415	not flagged	no
sample_18	ASD	high	20.326	flagged	yes
sample_19	Control	–	9.258	not flagged	yes
sample_20	Control	–	11.349	not flagged	yes

- Demo web application: upload a raw ROI time series and get the detector flag and a severity band. On 20 random held-out test files (10 control, 10 ASD) the output was right for 11 and the detector flagged 1.
- Files were drawn at random, never chosen by model output; the table shows every result, including the wrong ones.

DISCUSSION

WHAT WORKED

- Stable training with no representation collapse (embedding std 1.03–1.16).
- Generator works as designed: 7,200 records, 0 rule violations, severity-monotonic.
- Sub-structure edits detected 45.8% of the time vs 4.8% for random edits (9.6×).
- Leakage-safe protocol: patient-level split, thresholds from validation controls only.
- Complete pipeline plus a demo application and 20 sample scans.

CHALLENGES

- Detector is at chance on ABIDE I: AUROC 0.453, accuracy 53.4%.
- A single step-level threshold flagged all 131 test patients; fixed with a patient-level threshold.
- Few-shot F1 gains come from flagging more patients; AUROC stays about 0.50.
- Severity estimate is below a majority guess (43.3% vs 55.0%).
- Only 20 of 360 regions, one training run, no site-level test.

INTERPRETATION

- A diagnosis label is a weak sign of unusual connectivity dynamics; ASD is highly varied.
- Site effects and short, noisy windows may hide a real signal.
- Detectability of generated anomalies is self-referential: the same model guides and scores the edits.
- Guided and random generation behave alike in the few-shot test.
- Research prototype: no clinical claim.

CONCLUSION

- Built and tested an end-to-end pipeline: time-resolved brain graphs, brain-region subgraph tracking, temporal anomaly scoring, synthetic anomaly generation, few-shot adaptation and a severity output.
- All six objectives were completed, including a working demo web application.
- The anomaly generator works as designed: 0 rule violations, and 45.8% of sub-structure edits are detected against 4.8% for random edits.
- On ABIDE I the detector is at chance (AUROC 0.453, accuracy 53.4%); few-shot F1 gains come from a higher flag rate (AUROC about 0.50).
- Severity estimation is below a majority guess (43.3% vs 55.0%).
- Research prototype only: no clinical claim, and all results are reported as found.

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